

Type	Code	n_d	ν_d	$n_F - n_C$	n_e	ν_e	n_t	n_r	n_C	n_D	n_F	n_g
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CF	1	540-510	1.54041	50.95	0.010607	1.54294	50.68	1.52889	1.53541	1.53722	1.54032	1.54782	1.55374
	2	526-511	1.52630	51.05	0.010310	1.52875	50.78	1.51508	1.52144	1.52319	1.52621	1.53350	1.53922
	3	515-546	1.51454	54.63	0.009419	1.51678	54.36	1.50409	1.51007	1.51169	1.51446	1.52111	1.52630
	4	534-515	1.53358	51.54	0.010352	1.53604	51.26	1.52229	1.52870	1.53046	1.53349	1.54081	1.54658
	5	523-510	1.52310	50.95	0.010267	1.52554	50.67	1.51187	1.51826	1.52001	1.52301	1.53028	1.53599
	6	517-522	1.51742	52.15	0.009921	1.51978	51.88	1.50651	1.51273	1.51443	1.51733	1.52435	1.52985

n_h	n_i	Specific Gravity	Color	Bubble Group	Chemical Properties		Thermal Properties			Mechanical Properties		Remarks	Custom-ary Type
					D _w	D _A	T _g	T _s	α	H _k	F _A		

	1.55875	1.56745	2.78	34/31	2	2	1	475	525	107	490(5)	120		KF	1
	1.54403	1.55241	2.72	34/30	2	3	1	435	495	98	490(5)	110			2
	1.53066	1.53818	2.57	33/30	2	3	1	460	500	87	495(5)	110			3
	1.55143	1.55983	2.78	33/31	2	4	1	430	470	98	490(5)	110			4
	1.54082	1.54921	2.71	33/30	2	3	1	470	535	73	485(5)	110			5
	1.53449	1.54254	2.67	33/30	2	2	1	440	500	77	500(5)	100			6

Type	Code	n_d	ν_d	$n_F - n_C$	n_e	ν_e	n_t	n_r	n_C	n_D	n_F	n_g
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SbF	1	551-495	1.55115	49.52	0.011131	1.55380	49.26	1.53894	1.54590	1.54780	1.55105	1.55893	1.56514
	2	529-516	1.52944	51.64	0.010252	1.53188	51.40	1.51794	1.52458	1.52634	1.52935	1.53659	1.54225
	5	521-528	1.52130	52.78	0.009877	1.52365	52.55	1.51014	1.51659	1.51830	1.52121	1.52818	1.53359
	6	527-511	1.52682	51.08	0.010313	1.52928	50.85	1.51526	1.52193	1.52370	1.52673	1.53401	1.53970
BaFL	2	571-509	1.57099	50.86	0.011227	1.57366	50.56	1.55898	1.56573	1.56762	1.57089	1.57885	1.58511
	3	571-530	1.57135	52.98	0.010784	1.57392	52.69	1.55971	1.56628	1.56810	1.57125	1.57889	1.58485
	4	580-537	1.57957	53.71	0.010792	1.58214	53.42	1.56784	1.57448	1.57632	1.57948	1.58711	1.59306
	6	589-529	1.58904	52.93	0.011128	1.59169	52.64	1.57696	1.58382	1.58569	1.58894	1.59682	1.60294

n_h	n_i	Specific Gravity	Color	Bubble Group	Chemical Properties		Thermal Properties				Mechanical Properties		Remarks	Custom-ary Type
					D_w	D_A	T_g	T_s	α		H_k	F_A		

1.57039	1.57953	2.72	37/32	1	4	3	490	550	69		450(5)	120		KzF 1
1.54699	1.55520	2.57	35/30	1	5	4	460	530	58		470(5)	110		2
1.53812	1.54596	2.49	35/30	1	6	5	445	520	61		440(4)	110		5
1.54447	1.55274	2.54	36/30	1	6	5	435	510	59		415(4)	120		6
1.59040	1.59956	3.19	34/31	2	2	1	535	595	85		495(5)	140		BaLF 2
1.58985	1.59849	3.18	34/31	2-3	1	1	530	585	100		525(5)	140		3
1.59805	1.60669	3.23	37/33	2	2	2	580	640	78		525(5)	130		4
1.60803	1.61693	3.33	36/32	2-3	4	2	565	620	71		545(5)	130		6

KzF
BaLF

SbF
BaFL

Type	Code	n_d	ν_d	$n_F - n_C$	n_e	ν_e	n_t	n_r	n_C	n_D	n_F	n_g
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FEL	1	548-458	1.54814	45.82	0.011962	1.55098	45.54	1.53546	1.54257	1.54457	1.54803	1.55653	1.56328
	2	541-472	1.54072	47.20	0.011457	1.54344	46.92	1.52846	1.53536	1.53729	1.54062	1.54875	1.55519
	3	560-471	1.56013	47.09	0.011895	1.56295	46.79	1.54745	1.55460	1.55658	1.56002	1.56848	1.57519
	4	561-452	1.56138	45.23	0.012413	1.56433	44.94	1.54825	1.55561	1.55768	1.56127	1.57009	1.57713
	6	532-488	1.53172	48.84	0.010888	1.53431	48.57	1.52000	1.52661	1.52845	1.53162	1.53934	1.54540
	7	549-454	1.54883	45.44	0.012078	1.55170	45.16	1.53604	1.54321	1.54523	1.54872	1.55731	1.56413

n_h	n_i	Specific Gravity	Color	Bubble Group	Chemical Properties		Thermal Properties			Mechanical Properties		Remarks	Custom-ary Type
					D _w	D _A	T _g	T _s	α	H _k	F _A		

	1.56902	1.57919	2.95	35/31	1	3	1	430	475	99	475(5)	120		LLF 1
	1.56068	1.57033	2.86	34/31	1	3	1	435	480	91	455(5)	120		2
	1.58089	1.59096	3.01	34/31	1	2	1	510	570	77	545(5)	120		3
	1.58313	1.59372	3.02	34/31	1-2	3	1	450	500	99	500(5)	120		4
	1.55054	1.55957	2.79	33/30	2	3	1	430	490	85	495(5)	120		6
	1.56994	1.58023	2.98	34/31	1	3	1	415	480	89	475(5)	120		7

Type	Code	n_d	ν_d	$n_F - n_C$	n_e	ν_e	n_t	n_r	n_C	n_D	n_F	n_g
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BaF	3	583-465	1.58267	46.46	0.012541	1.58565	46.18	1.56956	1.57684	1.57893	1.58256	1.59147	1.59851
	4	606-439	1.60562	43.88	0.013803	1.60890	43.57	1.59128	1.59926	1.60153	1.60550	1.61533	1.62321
	5	607-493	1.60729	49.34	0.012309	1.61022	49.03	1.59418	1.60154	1.60361	1.60718	1.61592	1.62284
	6	589-485	1.58900	48.54	0.012134	1.59189	48.25	1.57607	1.58333	1.58537	1.58890	1.59750	1.60434
	7	608-462	1.60801	46.21	0.013157	1.61114	45.92	1.59418	1.60190	1.60409	1.60790	1.61725	1.62470
	8	624-471	1.62374	47.05	0.013256	1.62689	46.77	1.60977	1.61756	1.61978	1.62362	1.63303	1.64051
	9	643-479	1.64328	47.94	0.013418	1.64647	47.64	1.62904	1.63703	1.63927	1.64316	1.65269	1.66024
	10	670-472	1.67003	47.20	0.014197	1.67340	46.90	1.65486	1.66342	1.66579	1.66990	1.67999	1.68799
	11	667-483	1.66672	48.30	0.013805	1.67000	47.97	1.65213	1.66033	1.66262	1.66660	1.67642	1.68420
	12	639-451	1.63930	45.07	0.014185	1.64267	44.77	1.62451	1.63274	1.63509	1.63918	1.64927	1.65734

n_h	n_i	Specific Gravity	Color	Bubble Group	Chemical Properties		Thermal Properties			Mechanical Properties		Remarks	Custom-ary Type
					D _w	D _A	T _g	T _s	α	H _k	F _A		

	1.60448	1.61508	3.28	35/32	2	2	1	515	555	95	485(5)	160		BaF 3
	1.62994	1.64188	3.50	36/32	1	3	1	550	580	84	475(5)	160		4
	1.62872	1.63895	3.52	37/32	2	3	2	565	605	75	540(5)	140		5
	1.61014	1.62026	3.35	35/32	2	2	1	560	615	73	500(5)	140		6
	1.63106	1.64226	3.50	36/32	2	3	2	510	565	90	530(5)	170		7
	1.64687	1.65803	3.66	37/33	1-2	2	4	565	610	79	535(5)	160		8
	1.66664	1.67788	3.83	38/33	1	5	4	610	660	64	520(5)	150		9
	1.69480	1.70680	3.84	40/35	1	3	4	625	660	88	545(5)	190		10
	1.69077	1.70233	3.81	40/35	1	2	3	625	670	85	535(5)	180		11
	1.66424	1.67654	3.82	38/34	1	2	3	565	620	88	525(5)	170		12

Type	Code	n_D	ν_D	$n_F - n_C$	n_e	ν_e	n_t	n_r	n_C	n_D	n_F	n_g
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FL	1	573-426	1.57309	42.59	0.013457	1.57629	42.28	1.55907	1.56692	1.56912	1.57297	1.58258	1.59028
	2	589-410	1.58921	40.96	0.014384	1.59263	40.68	1.57431	1.58260	1.58496	1.58909	1.59934	1.60760
	3	582-420	1.58215	42.03	0.013852	1.58544	41.74	1.56773	1.57577	1.57805	1.58203	1.59190	1.59984
	4	578-417	1.57845	41.71	0.013869	1.58174	41.44	1.56397	1.57204	1.57433	1.57833	1.58820	1.59612
	5	581-409	1.58144	40.89	0.014221	1.58481	40.61	1.56673	1.57489	1.57723	1.58131	1.59145	1.59962
	6	567-428	1.56732	42.84	0.013242	1.57047	42.57	1.55342	1.56119	1.56339	1.56721	1.57663	1.58419
	7	575-415	1.57501	41.51	0.013853	1.57830	41.24	1.56065	1.56861	1.57090	1.57489	1.58475	1.59269
F	1	626-357	1.62588	35.74	0.017510	1.63003	35.49	1.60823	1.61791	1.62074	1.62573	1.63825	1.64845
	2	620-363	1.62004	36.30	0.017080	1.62409	36.05	1.60280	1.61225	1.61502	1.61989	1.63210	1.64206

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n_h	n_i	Specific Gravity	Color	Bubble Group	Chemical Properties		Thermal Properties			Mechanical Properties		Remarks	Custom-ary Type
					D _w	D _A	T _g	T _s	α	H _k	F _A		

	1.59687	1.60863	3.16	34/31	1	3	1	435	480	101	440(4)	150		LF	1
	1.61468	1.62738	3.30	36/32	1	3	1	435	485	99	445(4)	150			2
	1.60666	1.61884	3.22	35/32	1	2	1	460	505	93	465(5)	150			3
	1.60291	1.61510	3.20	35/31	1	4	1	440	485	93	465(5)	140			4
	1.60664	1.61917	3.24	34/31	1	3	1	420	480	98	445(4)	140			5
	1.59067	1.60221	3.12	34/31	1	2	1	450	495	84	470(5)	130			6
	1.59950	1.61166	3.20	34/31	1	3	1	435	475	89	450(5)	140			7
	1.65730	1.67343	3.65	36/33	1	3	1	440	480	87	435(4)	170		F	1
	1.65070	1.66631	3.60	36/32	1	3	1	435	480	101	425(4)	170			2

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Type	Code	n_d	ν_d	$n_F - n_C$	n_e	ν_e	n_t	n_r	n_C	n_D	n_F	n_g
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F	3	613-370	1.61293	36.96	0.016582	1.61686	36.70	1.59610	1.60535	1.60805	1.61278	1.62463	1.63428
	4	617-366	1.61659	36.61	0.016840	1.62058	36.35	1.59955	1.60891	1.61164	1.61644	1.62848	1.63829
	5	603-380	1.60342	38.01	0.015873	1.60718	37.75	1.58727	1.59615	1.59874	1.60328	1.61461	1.62380
	6	636-353	1.63636	35.34	0.018007	1.64062	35.08	1.61830	1.62818	1.63108	1.63620	1.64909	1.65960
	7	625-356	1.62536	35.58	0.017575	1.62953	35.32	1.60774	1.61739	1.62022	1.62521	1.63779	1.64807
	8	596-392	1.59551	39.22	0.015184	1.59912	38.96	1.57992	1.58854	1.59103	1.59538	1.60621	1.61498
	9	620-381	1.62045	38.09	0.016289	1.62431	37.83	1.60372	1.61298	1.61564	1.62030	1.63193	1.64138
	11	621-360	1.62096	35.95	0.017275	1.62505	35.68	1.60332	1.61309	1.61590	1.62081	1.63317	1.64334
	15	606-379	1.60565	37.90	0.015981	1.60944	37.64	1.58935	1.59833	1.60094	1.60551	1.61692	1.62618

n _h	n _i	Specific Gravity	Color	Bubble Group	Chemical Properties		Thermal Properties			Mechanical Properties		Remarks	Custom-ary Type
					D _w	D _A	T _g	T _s	α	H _k	F _A		

1.64264	1.65775	3.54	36/32	1	3	1	430	480	90	440(4)	170		F	3
1.64680	1.66218	3.57	36/32	1	3	1	435	480	97	430(4)	160			4
1.63174	1.64602	3.45	36/32	1	3	1	425	470	96	445(4)	160			5
1.66874	1.68538	3.74	38/33	1	2	2	440	485	90	435(4)	170			6
1.65701	1.67327	3.66	36/32	1	3	1	430	480	89	415(4)	160			7
1.62255	1.63612	3.37	36/32	1	3	1	440	480	90	450(5)	150			8
1.64956	1.66435	3.56	37/33	1	3	2	465	505	85	435(4)	150			9
1.65231	1.66901	2.67	40/36	1	1	1	575	620	90	620(6)	140			11
1.63419	1.64861	3.45	36/31	1	3	1	425	480	98	425(4)	140			15

Type	Code	n_d	ν_d	$n_F - n_C$	n_e	ν_e	n_t	n_r	n_C	n_D	n_F	n_g
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BaFD I	626-391	1.62606	39.09	0.016018	1.62986	38.80	1.60989	1.61874	1.62134	1.62592	1.63736	1.64662
2	664-359	1.66446	35.89	0.018515	1.66884	35.62	1.64601	1.65605	1.65903	1.66430	1.67755	1.68838
3	607-404	1.60717	40.36	0.015044	1.61074	40.08	1.59176	1.60025	1.60272	1.60704	1.61776	1.62644
4	651-383	1.65128	38.31	0.016999	1.65531	38.03	1.63405	1.64352	1.64628	1.65113	1.66328	1.67315
5	603-424	1.60323	42.39	0.014230	1.60661	42.10	1.58847	1.59668	1.59902	1.60311	1.61325	1.62141
6	668-419	1.66755	41.93	0.015920	1.67133	41.64	1.65115	1.66022	1.66284	1.66741	1.67876	1.68791
7	702-412	1.70154	41.15	0.017049	1.70559	40.86	1.68408	1.69372	1.69651	1.70139	1.71356	1.72339
8	723-380	1.72342	37.99	0.019041	1.72793	37.72	1.70410	1.71471	1.71781	1.72325	1.73685	1.74793
10	650-393	1.65016	39.34	0.016526	1.65408	39.07	1.63338	1.64259	1.64528	1.65001	1.66183	1.67136
14	700-349	1.69968	34.92	0.020034	1.70442	34.66	1.67965	1.69059	1.69382	1.69950	1.71385	1.72565

n_h	n_i	Specific Gravity	Color	Bubble Group	Chemical Properties		Thermal Properties			Mechanical Properties		Remarks	Custom-ary Type
					D_w	D_A	T_g	T_s	α	H_k	F_A		

1.65461	1.66906	3.67	37/33	1	2	1	460	515	91	460(5)	180		BaSF I
1.69782	1.71499	4.02	39/34	2	2	2	485	540	85	455(5)	200		2
1.63393	1.64735	3.50	36/32	1-2	2	1	475	530	85	475(5)	150		3
1.68168	1.69702	3.80	39/35	1-2	2	1	500	545	86	485(5)	180		4
1.62841	1.64089	3.48	36/32	1-2	2	1	470	515	97	485(5)	170		5
1.69579	1.70986	3.79	40/35	1	3	3	565	625	83	530(5)	180		6
1.73189	1.74719	3.91	41/36	1	3	4	630	675	82	560(6)	190		7
1.75759	1.77526	3.93	41/37	1	3	4	585	650	83	550(6)	180		8
1.67961	1.69445	3.81	39/35	2	3	3	505	550	90	460(5)	150		10
1.73599	1.75501	3.98	41/36	3	2	4	525	560	82	480(5)	190		14

Type	Code	n_d	ν_d	$n_F - n_C$	n_e	ν_e	n_t	n_r	n_C	n_D	n_F	n_g
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FD	1	717-295	1.71736	29.50	0.024317	1.72311	29.27	1.69369	1.70649	1.71032	1.71715	1.73464	1.74920
	2	648-338	1.64769	33.84	0.019142	1.65223	33.59	1.62861	1.63902	1.64210	1.64753	1.66124	1.67251
	3	740-282	1.74000	28.24	0.026200	1.74619	28.03	1.71474	1.72831	1.73243	1.73977	1.75863	1.77439
	4	755-275	1.75520	27.53	0.027432	1.76168	27.32	1.72883	1.74299	1.74729	1.75496	1.77472	1.79126
	5	673-322	1.67270	32.17	0.020909	1.67765	31.93	1.65209	1.66327	1.66661	1.67252	1.68752	1.69989
	6	805-255	1.80518	25.46	0.031620	1.81265	25.27	1.77524	1.79120	1.79611	1.80491	1.82773	1.84701
	7	640-346	1.63980	34.57	0.018508	1.64419	34.31	1.62126	1.63141	1.63439	1.63964	1.65290	1.66376
	8	689-312	1.68893	31.16	0.022111	1.69416	30.93	1.66722	1.67898	1.68250	1.68874	1.70461	1.71775
	9	654-337	1.65446	33.72	0.019411	1.65906	33.47	1.63513	1.64567	1.64879	1.65429	1.66820	1.67961
	10	728-283	1.72825	28.32	0.025716	1.73432	28.10	1.70338	1.71677	1.72082	1.72803	1.74654	1.76208

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n_h	n_i	Specific Gravity	Color	Bubble Group	Chemical Properties		Thermal Properties			Mechanical Properties		Remarks	Custom-ary Type
					D_w	D_A	T_g	T_s	α	H_k	F_A		

1.76204	1.78584	4.46	40/35	1	3	3	430	475	88	390(4)	180	SF	1
1.68236	1.70035	3.85	36/32	1	3	2	440	485	93	410(4)	180		2
1.78837	1.81447	4.65	41/35	1	3	2	440	485	82	390(4)	200		3
1.80598	1.83375	4.78	42/35	1-2	3	4	435	480	85	380(4)	200	L(41/35)	4
1.71074	1.73064	4.08	38/34	1	3	2	425	470	88	410(4)	180		5
1.86430	1.89708	5.19	43/36	1	3	3	435	470	86	365(4)	230	L(42/36)	6
1.67323	1.69045	3.79	36/33	1	3	2	435	485	91	410(4)	190		7
1.72931	1.75057	4.21	39/34	1	3	3	425	470	77	400(4)	190	L(38/34)	8
1.68958	1.70778	3.88	37/33	1	2	1	450	500	87	430(4)	160		9
1.77597	—	4.29	42/37	1-2	2	1	460	490	85	430(4)	200	L(41/37)	10

63

SF

FD

FDS

FF

LaFL

LaF

MBF

TAF

MBFD

TAFD

TAFD

AUG

AUG

Type	Code	n_d	ν_d	$n_F - n_C$	n_e	ν_e	n_t	n_r	n_C	n_D	n_F	n_g
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FD	11	785-257	1.78472	25.70	0.030529	1.79193	25.49	1.75581	1.77128	1.77598	1.78445	1.80651	1.82519
	12	648-338	1.64831	33.79	0.019184	1.65285	33.54	1.62919	1.63963	1.64271	1.64814	1.66189	1.67319
	13	741-278	1.74077	27.76	0.026681	1.74706	27.55	1.71509	1.72885	1.73307	1.74054	1.75975	1.77597
	14	762-266	1.76182	26.55	0.028692	1.76859	26.34	1.73448	1.74908	1.75357	1.76158	1.78226	1.79981
	15	699-301	1.69895	30.05	0.023259	1.70444	29.81	1.67623	1.68854	1.69221	1.69874	1.71547	1.72941
	18	722-292	1.72151	29.24	0.024673	1.72734	29.02	1.69753	1.71046	1.71436	1.72129	1.73903	1.75381
	19	667-331	1.66680	33.06	0.020170	1.67157	32.81	1.64687	1.65770	1.66092	1.66662	1.68109	1.69299
	20	720-293	1.72022	29.30	0.024581	1.72603	29.08	1.69630	1.70920	1.71310	1.72001	1.73768	1.75244

n_h	n_i	Specific Gravity	Color	Bubble Group	Chemical Properties		Thermal Properties			Mechanical Properties		Remarks	Customary Type
					D _w	D _A	T _g	T _s	α	H _k	F _A		

1.84216	—	4.84	44/38	1-2	1	3	445	495	90	410(4)	260	L(43/38)	SF 11
1.68307	1.70116	3.74	38/35	1	3	1	440	490	95	415(4)	180		12
1.79046	—	4.51	42/37	1	2	2	455	505	88	410(4)	210	L(41/37)	13
1.81558	—	4.64	43/37	1-2	2	3	440	490	85	405(4)	240	L(42/37)	14
1.74177	—	4.17	41/37	1	1	1	445	490	82	395(4)	170	L(40/37)	15
1.76686	1.79110	4.50	40/35	1	3	3	435	480	85	380(4)	210		18
1.70340	1.72248	4.01	38/34	1	2	2	430	480	86	425(4)	170		19
1.76555	1.78986	4.48	40/35	1	3	3	445	485	87	370(4)	180		20